

Mani K Chettri

PhD Scholar in Physics

Department of Physics, School of Physical Sciences
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Research Interests

Kinetic Alfvén wave (KAW) turbulence and dissipation at ion and sub-ion scales; nonlinear KAW-ion-acoustic and KAW-magnetosonic coupling; pseudospectral and split-step Fourier simulations of nonlinear plasma models; multi-mission analysis of MMS, Wind, Parker Solar Probe, and Aditya-L1 observations; collisionless plasma instabilities.

Education

PhD in Physics, Sikkim University, Gangtok, India

2022 – present

Thesis title: Numerical Study of Nonlinear Plasma Phenomena in the Heliosphere

Supervisors: Dr. Hemam Dinesh Singh (NSUT, New Delhi) and Dr. Rupak Mukherjee (Sikkim University, IUCAA Visiting Associate)

Research: KAW turbulence in the solar wind and magnetosphere using pseudospectral simulations of modified Zakharov and NLSE models and analysis of MMS, Wind, Parker Solar Probe, and Aditya-L1 observations.

MSc in Physics, Sikkim Manipal Institute of Technology, Majitar

2020 – 2022

Dissertation: The Cosmic Microwave Background Radiation and the Big Bang

BSc Physics (Honours), Nar Bahadur Bhandari Degree College, Tadong

2016 – 2019

Publications

Peer-Reviewed Journal Articles

1. Shrivastav, V., **Chettri, M. K.**, Singh, H. D., Singh, B., and Mukherjee, R. (2026). Suppression of two-stream and Weibel instabilities in ultra-relativistic electron-positron plasma. *Journal of Subatomic Particles and Cosmology*, 100421. doi:10.1016/j.jspc.2026.100421
2. **Chettri, M. K.**, Shrivastav, V., Singh, B., Mukherjee, R., and Singh, H. D. (2026). Inertial-range suppression and ponderomotive density cavitation in broadband sub-Alfvénic turbulence under plasma sheet boundary layer conditions. *Journal of Subatomic Particles and Cosmology*, 100383. doi:10.1016/j.jspc.2026.100383 arXiv:2603.14436
3. Mukherjee, R., **Chettri, M. K.**, Shrivastav, V., Singh, B., and Singh, H. D. (2026). How big is the Sun? A measurement from India's Aditya-L1 spacecraft. *PANE Journal of Physics Education*, 1(1), 60–74. Article

4. Shrivastav, V., **Chettri, M. K.**, Singh, H. D., Singh, B., and Mukherjee, R. (2026). Weibel instability in collisionless plasmas across astrophysical and laboratory shocks. *Contributions to Plasma Physics*, Early View. doi:10.1002/ctpp.70147
5. **Chettri, M. K.**, Shrivastav, V., Mukherjee, R., Gaur, N., Sharma, R., and Singh, H. D. (2024). Nonlinear coupling of kinetic Alfvén waves and ion acoustic waves in the inner heliosphere. *Research in Astronomy and Astrophysics*, **24**(10), 105009. doi:10.1088/1674-4527/ad76ee

Preprints and Submitted Manuscripts

1. **Chettri, M. K.**, Mukherjee, R., and Singh, H. D. (2026). Electromagnetic signatures of kinetic Alfvén wave turbulence at ion inertial scales in Earth's high- β magnetosheath. *Submitted to Advances in Space Research*.
arXiv:2605.25199
2. **Chettri, M. K.**, Singh, H. D., and Mukherjee, R. (2026). MMS observations of kinetic Alfvén wave turbulence and steep kinetic-range spectra in the outer plasma sheet boundary layer. *Submitted to Journal of Atmospheric and Solar-Terrestrial Physics*.
arXiv:2603.07969 SSRN 6447726
3. **Chettri, M. K.**, Singh, H. D., Shrivastav, V., Singh, B., and Mukherjee, R. (2025). Damped kinetic Alfvén waves in Earth's magnetosheath: numerical simulations and MMS observations.
arXiv:2512.09828
4. **Chettri, M. K.**, Shrivastav, V., Singh, B., Mukherjee, R., and Singh, H. D. (2026). A validated analysis workflow for near-Earth spacecraft plasma observations: methodology, quality controls, and systematic error assessment.
SSRN 6394663
5. Shrivastav, V., **Chettri, M. K.**, Singh, B., Singh, H. D., and Mukherjee, R. (2026). Linear-wave bound on electromagnetic energy equipartition at sub-electron scales in non-relativistic plasmas.
arXiv:2604.18472
6. Shrivastav, V., **Chettri, M. K.**, Singh, H. D., Singh, B., and Mukherjee, R. (2026). Ion channel dynamics in temperature-dependent Weibel instability saturation.
arXiv:2604.21875
7. Shrivastav, V., **Chettri, M. K.**, Singh, H. D., and Mukherjee, R. (2025). Suppression of two-stream instability in relativistic electron-ion plasmas.
arXiv:2508.14362

Software

PLASMAform: Interactive Space Plasma Formulary

2025 – present

mkchettri.in/plasmaform, GitHub, MIT License

Browser-based research tool with 50 formulas in 10 categories, CGS/SI unit conversion, KAW diagnostics, instability thresholds, reconnection parameters, mission presets, and validation examples.

Conference Presentations

- 52nd EPS Conference on Plasma Physics (EPS-2026), poster** 29 Jun – 3 Jul 2026
Edinburgh International Conference Centre (EICC), Edinburgh, UK
Ponderomotive generation of density cavities from kinetic Alfvén wave–slow-mode coupling.
- National Conference on Trends in Science and Technology (TIST-2026), oral** 27 – 28 Feb 2026
Salesian College, Siliguri, India
Damped kinetic Alfvén waves in Earth’s magnetosheath: numerical simulations and MMS observations.
- 5th Conference on Plasma Simulation (CPS-2026), poster** Jan 2026
IPS Academy and IIT Indore, India
KAW turbulence simulations in the plasma sheet boundary layer with MMS comparison.
- XV PANE Conference 2025, oral** Dec 2025
Nagaland University, Lumami, India
Turbulent signatures of kinetic Alfvén waves in Earth’s plasma sheet boundary layer.

Selected Workshops and Professional Development

- 13th Aditya-L1 Support Cell Workshop** 6 – 15 Jul 2026
ARIES, Nainital, India. Training in Aditya-L1 data products and analysis.
- JPP Frontiers of Plasma Physics Colloquium, online** May – Jun 2026
Journal of Plasma Physics (Cambridge University Press). Colloquia: *A parallel-kinetic-perpendicular moment model for magnetised plasmas; Can Machine Learning Discover Closure Relations in Plasma Physics?*
- Elsevier Researcher Academy, Certificates of Excellence** May 2026
Research Ethics (5 hours 30 minutes) and Funding (4 hours 50 minutes).
- Training Programme on AI and ML in Space Exploration** 23 – 28 Feb 2026
India Space Academy (ISA), Viksit Bharat Abhiyan, online. Certificate No. ISA-1003033.
- Aditya-L1 Support Cell Workshop** Dec 2025
IIT Jammu, organised by ISRO.
Data analysis using Aditya-L1 instruments (ASPEX, PAPA, MAG) via ISSDC/PRADAN.

Teaching Experience

- Science and Mathematics Instructor, Ascend Coaching Institute, Gangtok** 2022 – present
- Science Teacher, St. Mary’s Convent School, West Sikkim** 2019 – 2020
- Mathematics Tutor, Grace Academy, Gangtok** 2017 – 2019

Professional Memberships

- American Physical Society (APS), Division of Plasma Physics** 2026 – 2027
- European Geosciences Union (EGU), Solar-Terrestrial Sciences Division** 2026
- European Physical Society (EPS), Individual Member** 2026

Technical Skills

Programming	Python, Fortran, MATLAB, JavaScript/HTML, $\text{\LaTeX}$
Scientific Libraries	NumPy, SciPy, Matplotlib, Astropy, PySPEDAS, mpi4py, mpi4py-fft
Simulation Techniques	Pseudospectral with 2/3 de-aliasing, split-step Fourier, fourth-order Runge–Kutta, MPI-parallelised solvers
Spacecraft Data	MMS (FGM, FPI, EDP), Wind (MFI, SWE), Parker Solar Probe, Aditya-L1 (SUIT, SOLEXS, MAG, PAPA); via PySPEDAS and ISSDC/PRADAN
HPC	MPI parallelisation, IUCAA HPC cluster (Pune)
Version Control	Git, GitHub

Languages

Nepali, Hindi, and English (read, speak, and write).

Leadership and Service

President, Convention on Climate Change, Namchi Degree College	2019
President, Physicist Meet, Manan Kendra, Gangtok	2018

References

Dr. Hemam Dinesh Singh

Associate Professor
Dept. of Applied Physics
Netaji Subhas University of Technology
New Delhi, India
PhD Supervisor

Dr. Rupak Mukherjee

Associate Professor
Dept. of Physics
Sikkim University
Gangtok, India
PhD Co-Supervisor

Full contact details and additional references available on request.